

JUPITER / CLOUD MOTION

From cloud images to winds

A pedagogical guide to the JIRAM approach
of Ingersoll and colleagues

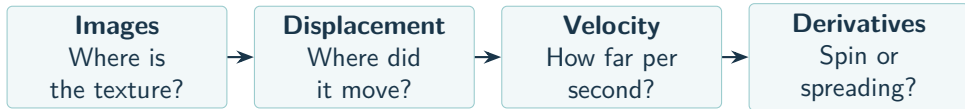
jiram_catalog science notes

9 September 2026

Ingersoll et al. (2022), *Nature Astronomy*.

Companion notes: `ingersoll_2022_tracking.tex`.

Follow one measurement through four steps



Every step needs something extra: geometry, correspondence, timing, then spatial error control.

What to remember

A plausible arrow is the beginning of a wind measurement.

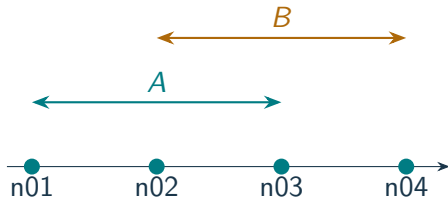
The experiment has a deliberately repeatable design

Input	JIRAM M-band, 2 February 2017; four sets of twelve images.
Mapping	SPICE/instrument geometry; polar orthographic map, 15 km pixels.
Motion	Tracker3 correlation; 15×15 pixel patches, 225 km sides.
Pair timing	n01–n03 and n02–n04; about 16 minutes, using actual image times.
Sampling	45 km vector spacing; native sampling varies from 22 to 14 km.
Quality	Low-entropy patches masked; 256 brightness bins per patch.
Diagnostics	Circulation/flux around 90, 180, 270, 360 km squares.

What to remember

A fixed recipe makes independent measurements comparable.

Four image sets give two separate measurements



- Comparable baselines.
- Nearby midpoints.
- No shared input image.

If the atmosphere changes little between estimates, their agreement can reveal a repeatable signal.

What to remember

Disjoint inputs remove one source of correlated measurement error.

Remove predicted camera motion first

Geometric mapping

Pixel $\rightarrow$ viewing ray $\rightarrow$ planetary intersection $\rightarrow$ common map.

Requires attitude, position, time, instrument model and reference surface.

Motion tracking

Compare texture after mapping.

Residual navigation error still looks like cloud motion.

Coordinate check

Image row, map y , and geographic north may point in different directions. A reflected coordinate basis can reverse vorticity's sign. Map derivatives require projection metrics or a quantified local approximation.

A ruler and a clock turn pixels into velocity

$$u = \frac{s_x d_x}{\Delta t}, \quad v = \frac{s_y d_y}{\Delta t}$$

Synthetic example

Grid: 20 km per pixel.

Time interval: 900 s.

Displacement: $(2.4, -1.2)$ pixels.

Result

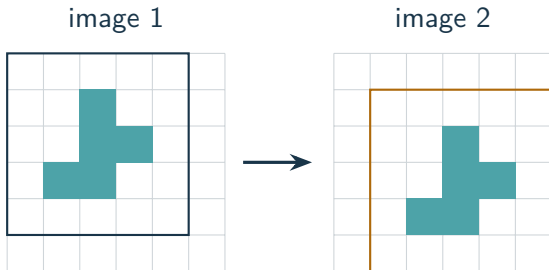
$(u, v) = (53.33, -26.67) \text{ m s}^{-1}$.

A 0.5-pixel relative navigation error alone contributes 11.11 m s^{-1} .

What to remember

Use each source pair's actual times. A mosaic is not necessarily an instant.

Find the same patch in the next image



1. Cut out a template.
2. Slide a matching window.
3. Score each shift.
4. Refine the best peak.

What to remember

The measurement uses an entire patch of texture.

Correlation compares patterns, with a validity mask

$$C(\mathbf{d}) = \frac{\sum_{i \in M(\mathbf{d})} (A_i - \bar{A})(B_i(\mathbf{d}) - \bar{B})}{\sqrt{\sum_{i \in M(\mathbf{d})} (A_i - \bar{A})^2 \sum_{i \in M(\mathbf{d})} (B_i(\mathbf{d}) - \bar{B})^2}}$$

- Means and sums use $M(\mathbf{d})$: pixels valid in both trial patches.
- Mean subtraction removes a constant brightness offset.
- Normalization removes a positive gain change.
- A local peak fit permits a subpixel displacement estimate.

What to remember

A gap filled with zero can become a convincing false feature.

Optical flow begins with a brightness assumption

$$I(\mathbf{x} + \mathbf{v}\Delta t, t + \Delta t) \simeq I(\mathbf{x}, t) \implies I_t + uI_x + vI_y \simeq 0.$$

Patch tracking

Search finite shifts of a neighborhood.

Differential optical flow

Fit brightness derivatives with spatial constraints.

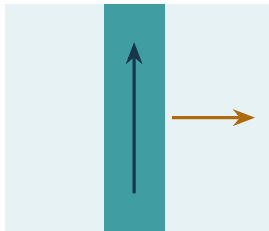
Both become ambiguous when texture evolves independently of motion:

$$I_t + \mathbf{v} \cdot \nabla I = S_I.$$

What to remember

Cloud formation or radiometric change can be mistaken for advection.

A straight stripe cannot reveal every motion component



Along the stripe:
locally invisible.

The brightness equation contains two unknown velocity components.

Texture with gradients in several directions supplies additional constraints.

A smooth or repetitive patch may have many plausible matches.

Texture quality needs more than one score

Histogram entropy

$$H = - \sum_{k:p_k>0} p_k \log_2 p_k$$

Uniform patch: low H .

Many intensity levels: higher H .

What it cannot guarantee

A smooth ramp can have high entropy.

Random noise can have high entropy.

Neither guarantees a unique, persistent match.

What to remember

Also inspect peak ambiguity, valid area, directional texture and forward/backward consistency.

Sampling, support and resolution answer different questions

Native sampling	How far apart are detector samples on Jupiter?
Map spacing	How fine is the resampled coordinate grid?
Template support	How large a patch constrains one velocity?
Vector spacing	How far apart do we report estimates?
Derivative scale	Over what area do we measure spin or spreading?

Interpolating an image or drawing more arrows does not add observations.

What to remember

The interface should display these scales separately.

Many arrows can reuse almost the same information



Overlapping templates
share texture.

Synthetic example:

Useful patch width: 240 km.

Vector spacing: 40 km $\rightarrow$ 10 km.

Sixteen times as many vectors per unit
area; no new photons.

What to remember

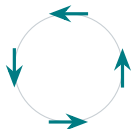
Do not count overlapping vectors as independent samples.

Vorticity measures spin; divergence measures spreading

Rigid rotation

$$(u, v) = (-\omega y, \omega x)$$

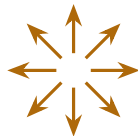
$$\zeta = v_x - u_y = 2\omega, \quad D = 0$$



Isotropic expansion

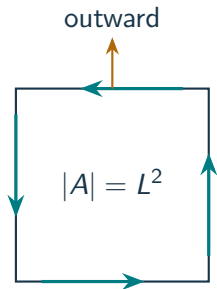
$$(u, v) = (\alpha x, \alpha y)$$

$$\zeta = 0, \quad D = u_x + v_y = 2\alpha$$



Both diagnostics have units s^{-1} ; uniform translation gives zero for both.

Measure a region through its boundary



Circulation / area

$$\bar{\zeta}_A = \frac{1}{|A|} \oint (u \, dx + v \, dy)$$

Outward flux / area

$$\bar{D}_A = \frac{1}{|A|} \oint (u \, dy - v \, dx)$$

Counterclockwise contour; physical Cartesian basis.

What to remember

These are area means. Missing edges must not silently become zero flow.

Small velocity errors can overwhelm a derivative

$$\sigma_{\bar{D}} \simeq \sigma_{\bar{\zeta}} \simeq \frac{2\sigma_e}{L}$$

Illustrative assumptions

Four independent edge means.

Each uncertain by 6.29 m s^{-1} .

Square side $L = 200 \text{ km}$.

Result

Derivative noise:

$$6.29 \times 10^{-5} \text{ s}^{-1}$$

Larger than a true divergence of
 $2 \times 10^{-5} \text{ s}^{-1}$.

What to remember

Real propagation must retain covariance: $\text{Var}(q) = \mathbf{w}^T \mathbf{C} \mathbf{w}$.

Agreement between repeats estimates a common signal

$$A = s + e_A, \quad B = s + e_B$$

Assume the same physical signal, independent errors, no signal–error covariance, and equal individual noise variances.

$$\text{Var}(A) = \text{Var}(B) = \sigma_s^2 + \sigma_n^2,$$

$$\text{Cov}(A, B) = \sigma_s^2,$$

$$\eta = \text{corr}(A, B) = \frac{\sigma_s^2}{\sigma_s^2 + \sigma_n^2}.$$

What to remember

Under this model, correlation is a signal fraction of variance.

Try the noise calculation with simple numbers

Each repeated measurement has standard deviation 10.

Their correlation is 0.36.

Common signal

$$\sigma_s^2 = 0.36(10^2)$$

$$\sigma_s = 6$$

Individual noise

$$\sigma_n^2 = 100 - 36$$

$$\sigma_n = 8$$

Noise after averaging

$$\sigma_{n,\text{mean}} = 8/\sqrt{2}$$

$$\sigma_{n,\text{mean}} = 5.66$$

What to remember

Correlation 0.36 does not mean 36% of vectors are correct.

A repeatable error can look like repeatable atmosphere

- **Shared navigation:** both estimates can inherit the same bias.
- **Changing clouds:** real evolution can lower repeat correlation.
- **Repeated texture failure:** the same ambiguity can fool both pairs.
- **Shared image:** adjacent pairs reuse a position error with opposite signs.

For independent position errors with equal variance σ_p^2 :

$$e_{12} = \frac{p_2 - p_1}{\Delta t}, \quad e_{23} = \frac{p_3 - p_2}{\Delta t} \Rightarrow \text{Cov}(e_{12}, e_{23}) = -\frac{\sigma_p^2}{\Delta t^2}.$$

What to remember

Disjoint pairs help; independent physical validation still matters.

The repeat measurements distinguish stronger and weaker signals

At a 180 km contour scale	Vorticity	Divergence
Repeat correlation	≈ 0.729	≈ 0.299

The fitted velocity uncertainty was approximately 7.8 m s^{-1} per component.

Under the repeat-measurement model, divergence contains a smaller signal fraction than vorticity.

What to remember

Velocity, vorticity and divergence need separate confidence assessments.

A shielding ring does not require reversed wind arrows

$$\zeta(r) = \frac{1}{r} \frac{d(rv_\theta)}{dr}$$

Consider a synthetic profile:

Its vorticity is

$$v_\theta = Cr^{-p}, \quad C > 0.$$

$$\zeta = (1 - p)Cr^{-p-1}.$$

The azimuthal wind remains positive.

For $p > 1$, the vorticity is negative.

What to remember

Wind decreasing faster than $1/r$ can produce an anticyclonic annulus.

Divergence does not directly measure vertical velocity

Simple incompressible continuity:
$$D = -\frac{\partial w}{\partial z}.$$

This constrains a vertical gradient. A compressible atmosphere requires density and vertical-structure information.

Ingersoll et al. did not detect the proposed divergence–anticyclonic vorticity correlation at their resolved scales.

Siegelman et al. used infrared optical-depth anomalies and additional surface-quasigeostrophic assumptions to investigate smaller scales.

What to remember

A direct wind diagnostic and a model-assisted reconstruction have different assumptions.

Our existing tracker is a starting point, with substantial tails

Comparison input	Median error	90th percentile	Within 1 px
Published maps	0.224 px	4.759 px	83.4%
Our reprojections	0.434 px	4.685 px	80.1%

24 same-letter tile pairs, subsampled reference positions.

Median vector-difference magnitudes: 3.45 and 6.69 m s^{-1} .

These compare with published vectors, not exact atmospheric truth.

What to remember

Good median agreement does not establish derivative accuracy.

Resolve three provenance details before claiming reproduction

1. **Software identity:** paper says Tracker3; local table history says TRACKER4. Equivalence is unverified.
2. **Sampling:** the inspected raw table has GRID=1 and 15 km map pixels. Its placement differs from the reported analysis grid. “45 km” is not evidence of cloud altitude.
3. **Coordinates:** sample/line and map/array axes differ. Verify the physical basis before assigning wind directions or curl signs.

What to remember

The GUI currently loads associated vector products; it does not yet supply this complete wind workflow.

Build the next features around inspectable measurements

1. Design the experiment

Four-set selector.
Exact source times.
Common valid coverage.
Instrument eligibility.

2. Inspect every vector

Paired patches.
Correlation surface.
Masks and uncertainty.
Saved source association.

3. Test the science

Contour-scale control.
Repeatability panel.
Synthetic benchmarks.
Reference comparisons.

What to remember

Start with a reproducible PJ4 benchmark; then validate other passes and JunoCam.

JunoCam needs its own motion-validation evidence

- Retain instrument-failure exclusions and native-band provenance.
- Preserve framelet timing, map support and source identity.
- Quantify navigation and band-registration errors.
- Check reflected-light changes separately from JIRAM thermal texture.

What to remember

A regular tensor, clean image or successful export does not establish accurate winds.

Check your understanding before building anything

1. Does halving vector spacing necessarily resolve smaller eddies?
2. Can a constant navigation translation create curl in an ideal plane?
3. Can two biased estimates correlate almost perfectly?
4. Does horizontal divergence uniquely determine upward velocity?

Answers: No. No—spatially varying errors are different. Yes. No—vertical structure is needed.

What to remember

Resolution, repeatability and physical accuracy are separate questions.

Sources and companion notes

Main paper

Ingersoll et al. (2022), *Vorticity and divergence at scales down to 200 km within and around the polar cyclones of Jupiter*.

Full coauthor-hosted PDF: Methods pp. 1284–1285, Figs. 2–3, Table 1.

Related dynamical inference

Siegelman et al. (2022), Moist convection drives an upscale energy transfer at Jovian high latitudes.

Repository and full derivations

Repository PJ4 validation; tracking.py; api/science.py.

Companion: docs/pedagogy/ingersoll_2022_tracking.tex.

Original diagrams and synthetic examples; no article figures reproduced.

Judgment calls and what remains unverified

- Chose a short measurement tutorial, with full derivations in companion notes, rather than reproducing the complete vortex-dynamics analysis.
- Kept the published recipe, local implementation, original teaching examples and proposed features distinct.
- Treated reference agreement as a benchmark, not perfect wind truth; raw placement and analysis-grid sampling remain separate.
- Supplementary methods were inaccessible: the entropy cutoff, exact quadrature, raw-to-analysis aggregation and Tracker version equivalence remain unverified. No settings were invented to fill those gaps.

What to remember

Next implementation requires an accepted specification and scientific validation.